

INSTITUTIONS AS FAMILIES OF GAME FORMS*

By LEONID HURWICZ

University of Minnesota

1. Introduction

Recent decades have witnessed a revival of interest in the role of institutions in economic processes. The importance attached to the work of Coase, North, Williamson and others attests to this. The works of these authors contain illustrative examples of institutions: property rights (common or private property), the state as guarantor of rights, law, markets (competitive, oligopolistic, monopolistic), money (commodity or fiat money), liability for damage by smoke, state regulation of economic activities (e.g., price controls, anti-trust laws), various forms of land tenure (sharecropping, renting, utilization of wage-labour), various forms of business organization (corporations, partnerships, non-profits), company towns, franchise systems, and many others – including even the week, viewed as an institution.

Specific institutions have been studied with the help of familiar analytical tools: demand and cost functions, principal-agent, and game-theoretical models. The comparison of sharecropping (*métayage* or on “shares”) system with renting (the “English” system) is already found in Marshall’s 1890 edition of his *Principles of Economics* (Book VII, Ch. XI, §§ 4, 5, 6, pp. 682–689). But serious analytic study of institutions as a general phenomenon calls for a rigorous definition of the concept of an institution within the framework of a formal model. Ideally, the definition should fit most if not all of the examples such as those above.

Not all those stressing the importance of institutions have attempted to provide such a definition. Those who did, as pointed out by Ostrom (1986), had to face the multiplicity of meanings of the term and to make choices among them.

2. Institutions and game forms

2.1 Meanings A and B of the term ‘institution’

For North (1990, pp. 3, 4), who views institutions as rules of the game in a society, a basic dichotomy is between the meaning of institutions as such rules and the players whom he calls “organizations”. But these players are often also called institutions. They are entities or bodies, usually mission oriented and treated as juridical persons. Examples abound: a university, a church, a country’s legislative body. Although most “player”-institutions involve many (physical) persons and can be classified as organizations, there are also one-person “player”-institutions, as for instance a

* The present paper is an expanded version of my comments at the First Decentralization Conference in Japan, held at Keio University in November 1994. Due to its expository nature, it overlaps parts of some of my earlier papers cited in the References. Thanks are due to Professors Tatsuyoshi Saijo and Takehiko Yamato for numerous helpful suggestions. Responsibility, as always, remains the author’s.

country's presidency or an agency's ombudsman.¹⁾ We shall refer to "player"-institutions as the *A-institutions*.

The other meaning of 'institution', to be referred as the B meaning, and often called an *institutional arrangement*, is a matter of some controversy. As noted above, for North and many others²⁾ it consists of rules (in the language of game theory, *game forms*). For others, in particular Schotter (1981, p. 155), social³⁾ institutions are regularities or standards of behaviour.⁴⁾ In the language of the game theory, they are for him the *solutions* rather than rules of the game.⁵⁾

In what follows I shall be dealing primarily with the B meaning of the term "institution", i.e., with institutional arrangements. Hence in what follows "institution" is to be interpreted as synonymous with institutional arrangement (a *B-institution*), unless the contrary is specified.

As for a formal definition of a B-institution, mine will be close to the "rules (game forms)" interpretation, but when an institution's genesis is taken into account it will incorporate some of the regularities approach as well. To make this clear, some formal apparatus is needed.

2.2 Notation for games in normal form

In order to be clear about the formalization of the notion of rules of the game, we must be careful about distinguishing between a *game* and a *game-form*.⁶⁾ Games, as is well known, have different forms of representation, in particular the so-called *extensive form* specifying the individual moves and their consequences, and the *normal form* formulated in terms of admissible *strategies* and the outcomes they produce.

We shall mostly be dealing with the normal form, assuming a finite number n of players. The term "game" in its technical sense is applied to the ordered pair consisting of two n -tuples: that of *individual admissible strategy domains*, $S^1, \dots, S^n$, and that of (real-valued) payoff functions, $\pi^1, \dots, \pi^n$. Denote by $S = S^1 \times \dots \times S^n$, the (*joint*) *strategy space*, i.e., the Cartesian product of the individual domains, and by $N = \{1, \dots, n\}$ the set of players. For all i in N , the i -th payoff function has the (joint) strategy space S as its domain and takes its values in the reals. Let $s = (s^1, \dots, s^n)$, s in S , s^i in S^i for all i in N , be the n -tuple of strategies adopted by the players. The result is that the i -th player obtains the payoff $\pi^i(s)$, usually interpreted as the utility derived by the i -th player when s prevails. It is its payoff that each player is trying to maximize.

1) In one-person player-institutions one must distinguish between the player-institution and its incumbent.

2) In particular Riker (1982, p. 4) and Ostrom (1986). For an approach synthesizing the different concepts see Crawford and Ostrom (1994)

3) Although North, unlike Schotter, does not use the term 'social', he speaks of the rules of the game in *society*, so the intent seems the same.

4) It is interesting to note that von Neumann and Morgenstern (1944, pp. 41, 512) use the term "standards of behavior" as representing the solution of a game. On the other hand, when using the term 'institutions' (as on p. 225, footnote 3), identified with forms of social organization (the example being *laissez-faire*), they seem to be referring to rules of the game rather than to solutions.

5) In (1989, pp. 50–1), Schotter speaks of institutions of type I (rules) and of type II (regularities of behaviour).

6) Although this is not customary, we sometimes hyphenate the term "game-form" in order to avoid misunderstandings that might arise, e.g., when we speak of game-forms in extensive form.

Formally, a game Γ then is defined as the double n -tuple, $\Gamma = (S^1, \dots, S^n; \pi^1, \dots, \pi^n)$ where, for each i in N , $\pi^i: S \rightarrow \mathbb{R}$. (Here $\mathbb{R}$ represents the reals).

2.3 Game forms vs. games

Since the payoff values represent the players' utilities, they depend both on rules of the game and on the players' preferences. To represent this situation, we first introduce the notion of an *outcome space*, denoted by Z , whose elements are the (physical) outcomes of the strategic decisions. In economic environments, the generic element of Z might be a resource allocation; for example, in a pure-exchange economy with m goods, a point z of the outcome space is a vector with $m \cdot n$ real components z_j^i , where z_j^i is the total amount of good j held after trading by trader i , so that Z is a vector space of dimension $m \cdot n$. In a voting model, the outcome space might consist of the set of candidates. In a social game, Z might be the set of prizes that might be given to players. Just who gets which prize depends, of course, on the strategies chosen by the players. The relationship between the strategies chosen and the resulting outcomes is defined by the outcome function, denoted by h .

The *game form*⁷⁾ G is formally defined as $G = (S^1, \dots, S^n, h)$, or more briefly as (S, h) where S is the Cartesian product of the S^i 's and $h: S \rightarrow Z$.

To express the relationship between game forms and games, we introduce utility functions to represent preferences.⁸⁾ The i -th player's *utility function* ϕ^i associates a real number, say u^i , with the outcome (e.g., resource allocation⁹⁾ or candidate elected) resulting from the strategies s chosen by the players. I.e., $u^i = \phi^i(z)$ where $z = h(s)$ and so $u^i = \phi^i(h(s))$. Hence the real number u^i represents the i -th player's utility resulting when the strategy n -tuple s prevails, so that u^i is the value of the i -th payoff function associated with s . It follows that the payoff function is the composite of the utility function and the outcome function. This can be written as

$$\pi^i(s) = \phi^i(h(s)) \quad \text{for all } i \text{ in } N \text{ and all } s \text{ in } S,$$

or, more compactly, as $\pi^i = \phi^i \circ h$.

2.4 "Rules of the game" are represented by game forms, not by games

We have been somewhat pedantic in formalizing the concepts of a game and of a game form, because the distinction between the two is crucial for understanding the meaning of the "rules of the game". Since the players' preferences are not part of the rules, it is the game form, rather than the game, that corresponds to the intuitive notion of the "rules of the game".¹⁰⁾ Hence "game form" is the important concept in defining institutions in terms of rules.¹¹⁾

7) In the economic literature often called the *mechanism*.

8) At the cost of complicating exposition it is possible to avoid the assumption (implicitly made here) that the preferences can be represented by utility functions.

9) This formulation allows for the possibility of non-selfish preferences. When preferences are selfish, the i -th player's utility in fact depends only on the subvector or "bundle" $z^i = (z_1^i, \dots, z_m^i)$ of the resource allocation z .

10) On the other hand, since institutions will be defined as the results of human activities, not all aspects of the game-form are to be viewed as institutional.

11) I.e., corresponding to Schotter's "type"!

In the literature, the models under consideration frequently involve numerical outcomes – e.g., monetary payments to individual consumers. Thus an element of the outcome space is $z = (z^1, \dots, z^n)$ where z^i is the payment to person i (a real number). Further, suppose that every individual is selfish, so that i 's utility depends only on the i -th component of z , say $u^i = \phi_i(z) = \psi^i(z^i)$ where ψ^i is a strictly increasing function. Finally, suppose that every one is “risk-neutral”, i.e., that the ψ^i is a linear (or affine) function of the payment z^i . Then, without loss of generality, one may take the function ψ^i to be identity, so that $u^i = \phi_i(z) = \psi^i(z^i = z^i)$ for each person i . Let h denote the outcome function, so that $h(s) = (z^1, \dots, z^n)$, and define $h^i(s) = z^i$. Then the payoff function in this model is given by $\pi^i(s) = z^i = h^i(s)$, and the n -tuple $\pi(s) = (\pi^1, \dots, \pi^n)$ is identical with the outcome function $h(s) = (h^1(s), \dots, h^n(s))$. It is easy to see that in this case the distinction between the outcome function (the n -tuple of the h^i 's) and the i -th payoff function (identical with the i -th-component of the outcome function) does not seem very important. But, of course, this is a very special case.

2.5 Representation of institutions by game forms involves more than restrictions on strategy domains

When an institution is defined by restrictions on the individual strategy domains (the sets S^i) imposed by an authority, it is natural to call such an institution a *regulation*.¹²⁾ North's (1990, p. 3) initial definition of institutions appears to include little¹³⁾ beyond regulations so defined: “Institutions are the rules of the game in a society or, more formally, are the humanly devised constraints that shape human interaction.”

But the notion of (B-) institutions as constraints is, in my view, too narrow – for two reasons. First, of the two components of a game form (S, h) they only deal with the first (the strategy domains) and, even among the properties of strategy domains, they seem to cover only restrictions of the individual strategy domains while ignoring the possibility of expansions¹⁴⁾ of these domains by institutional arrangements. Second, it is difficult to see how certain important institutions (e.g., markets, social insurance) could be described just in terms of strategy domains but without reference to the outcome functions. It is natural to represent arrangements such as taxes, penalties, or subsidies as properties of outcome functions.¹⁵⁾

12) The concept of a regulated economy in Reiter and Hughes (1981) is broader: it includes not only “direct constraints on behaviour” (corresponding to our concept of regulation) but also parameters θ^i determining the agents' incentives (e.g., taxes or subsidies).

13) It is somewhat broader because it refers to constraints that are “humanly devised” – but not necessarily by an authority. This might, for instance, cover commodity money regarded as a (spontaneously developed) institution but not qualifying as regulation.

14) Crawford and Ostrom (1994, p. 3), by contrast, do refer to opportunities as well as to constraints.

15) Admittedly, some of the arrangements mentioned above—such as taxes (or subsidies)—can alternatively be viewed as restrictions on (or expansions of) strategy domains, although this representation might make analysis more complicated. But in many cases, including these examples, there is additional machinery involved, needed for the processing and transmission of information as well as for enforcement. Representation through outcome functions (as well as addition of artificial players such as, e.g., the Internal Revenue Service agency, an A-institution) seems the appropriate way to represent such institutional arrangements.

2.6 An institution is a class of game forms, not just a single game form

An examination of examples of any phenomenon typically called an institution (in the sense of rules) shows, however, that it would not be natural to identify it with a particular game form. For instance, the institution of sharecropping specifies that the crops will be divided so that the landlord gets a fraction, say k (with $0 < k < 1$) and the labourer $1 - k$.¹⁶⁾ Now when sharecropping is formulated as a game form, the value of the share parameter k (entering the outcome function) must be specified, say $k = 1/2$. But to qualify as the institution of sharecropping any fractional value of k is acceptable. Hence the institution of sharecropping corresponds not to a particular game-form, say $G_{1/2}$ (with k specified to equal $1/2$) but to a class of game forms, say $\mathbb{G} = \{G_k: 0 < k < 1\}$.¹⁷⁾

Another illustration of the same point is provided by price controls, more specifically price ceilings. In game form representation, these define the seller's strategy (price) domain as having a finite upper bound, say p^{i*} , determined by the price control agency, say $S^i = \{p^i: p^i \leq p^{i*}\}$, where the seller is the i -th agent and p^i his/her strategy variable. To specify a game form requires a particular numerical value of p^{i*} . Denote such a game form by $G_{p^{i*}}$, characterized by a specific value of p^{i*} in the above formula for S^i . One would naturally say that price controls as an institution prevail regardless of the numerical value of the ceiling p^{i*} . Hence, again, the institution of (ceiling) price controls corresponds to a *class of game forms*, say $\mathbb{G} = \{G_{p^{i*}}: p^{i*} \text{ ranging over (finite!) real numbers}\}$.¹⁸⁾

Membership in a class of objects may be considered as equivalent to having a certain attribute (or collection of attributes) distinguishing members from non-members. Hence defining an institution as a class of game forms may be viewed as equivalent with defining a institution as an attribute (or collection of attributes) of game forms.

We may refer to the game forms belonging to the class representing the same institution (i.e., sharing certain attributes) as "institutionally equivalent". Typically, this equivalence is based on similarity of the incentive structure it engenders for the participants, but there is a great deal of arbitrariness, sometimes depending on the issues under consideration. As an illustration, much of the analysis of incentive

16) The 1969 *Nouveau Petit Larousse en couleurs* specifies (under *métayage*) that k should not exceed a third. Actually, the term *métayer* comes from a Latin word meaning 'a half'. Thus there might have been a time when $k = 1/2$ was the only permissible value under *métayage*. But in the present context, all values strictly between zero and one qualify as 'sharecropping'. (See Hurwicz (1979a, pp. 123–147, and esp. pp. 140–144, the section entitled 'Incentive structures for unilateral maximization', for a simple model in which one-half turns out to be the best value of k from the landlord's point of view when the labourer's effort cannot be measured or observed.)

17) Following Stiglitz (1974), let the labourer's reward, a component of the outcome function, be written as $r = ky + b$ where y is the size of the crop (regarded here as the worker's strategy variable) and k and b are real-valued parameters. Different forms of land tenure are defined by different regions in the (k, b) -parameter space. Thus (pure) sharecropping is defined by the conditions $b = 0$ and $0 < k < 1$; renting by $k = 1, b < 0$; wage-labour by $k = 0, b > 0$. Thus, under renting, $-b$ is the rental fee, while under wage-labour b is the wage.

18) Although the frameworks are different (dealing with imputations or strategy spaces), it is of interest to note the multiplicity of imputations in a single von Neumann-Morgenstern solution and the multiplicity of Nash solutions (e.g., in a coordination game) corresponding to the same institution in Schotter's treatment.

structures of the alternative land tenure form reaches conclusions that are valid for all pure sharecropping game forms, regardless of the value of the share parameter k (or a), provided it is strictly between 0 and 1; similarly for all rental arrangements, regardless of the value of the parameter $b < 0$, and all wage-labour arrangements regardless of the value of $b > 0$. (See footnote 17 for notation.) Hence we speak of all pure sharecropping game forms, regardless of the value of k as institutionally equivalent, etc.

2.7 Conditional game forms (“humanly devised” aspects of game forms)

2.7.1 Institutions as correspondences from environment's to game forms

Quite often a game form has two aspects: (1) those that are behavioural in nature and are (potentially at least) subject to alternative design and embody the essence of the institutional arrangements, and (2) those that are considered as given, either because they are determined by the laws of physics or, more generally, by what we call the (economic) environment (in particular by the existing resource endowments, and the current state of technology). In such cases it would be incorrect to identify the institutional arrangements with all of the game form (S, h) , and therefore a more refined definition of an institution must be formulated.

One example of such a game form is seen in a mechanism proposed in Hurwicz, Maskin and Postlewaite (1995) for situations in which no player knows the other players' endowments. The rules of the game call for each player to announce his/her own endowment (as well as estimates of the other players' endowments). This announcement is permitted to be an understatement but not an overstatement. The prohibition against overstatement is enforced by the requirement that the player “put on the table” the resources claimed in the endowment. Assuming that borrowing is somehow ruled out, it is then physically impossible to exaggerate. But this means that a given player's strategy domain depends on his/her true endowment. (In the usual economic models the endowments, preferences, and technologies determine the environment.) Thus the i -th strategy domain S^i is a function of the true value, denoted by ω^{io} , of the i -th agent's endowment. Therefore, the i -th strategy domain is denoted more accurately by $S^i(\omega^{io})$. Since ω^{io} is a component of the i -th characteristic e^i , and, in turn, e^i is a component of the environment $e = (e^1, \dots, e^n)$, we see that the rules of the game specify the i -th strategy domain by a functional relation from the space E of environments into the space of strategy domains, hence into the space of game forms. The institutional aspect, here the result of design (“humanly devised”), is this functional relation, rather than the strategy domain itself.

An instance where the outcome function of a mechanism depends on the existing technology, hence on the environment e , is given below in Section 6. Here the rules of the game specify (via the “reward functions”) how players are rewarded given the output. With y denoting output, consider (as in footnote 17 – the Stiglitz model) the class of (the worker's) linear reward functions written $r^i = ky + b$. (However, unlike in footnote 17 and in Section 6, effort – not output – is considered the worker's strategy variable). As in footnote 17, special cases of the reward functions (i.e., of the parameters k and b) produce alternative institutional arrangements (e.g., sharecropping, wage labour, renting) and are “humanly devised”. The player's strategy may be his/her intensity of effort; this will determine the output via the production function. The outcome function, relating reward (outcome) to effort (strategy), has a “humanly

devised" aspect (the reward functions) and an aspect determined by the environment (the production function). Thus the institutional aspect is in this example represented by a functional relation from the space E of environments to the space of outcome functions, rather than the outcome function itself.

The preceding two examples illustrate the fact that institutional phenomena should, in general, be represented not as game forms but as relationships between game forms and the economic environment. But since we think of institutions as associated with *classes* of mechanisms, this relationship must be viewed as a correspondence rather than a function. Thus in the preceding example the institution of sharecropping involves the worker's reward function ("humanly devised") of the form $r^i = ky$ where y is the output and k a number between zero and one. But suppose the strategy variable s^i is the worker's effort x^i , so that $s^i = x^i$, and the output is determined by the production function $y = cx^i$, where c is the technologically determined productivity of effort parameter, considered as part of the environment and not institutional in nature. Then the worker's outcome function is $h^i(s) = dx^i = kcx^i$, involving both the institutional aspect (k) and the environment aspect (c). Thus it is not the outcome function h^i that represents the institution of sharecropping, but rather the correspondence that associates with each environment (productivity parameter c) is an outcome function $h^i(\cdot)$, where $h^i(s) = dx^i$ where d/c is a number between zero and one.

In formal notation we have the following situation. Let $\mathcal{G}$ denote the class of conceivable game forms. We may then consider an institution to be a correspondence, say Λ from the space of environments E into $\mathcal{G}$, i.e., $\Lambda: E \rightarrow \mathcal{G}$. Given an environment e , the resulting class of game forms, $\mathcal{G} = \Lambda(e)$, embodies not only institutions but also the characteristics of the environment; it is the correspondence Λ itself that represents the purely institutional aspect of the game forms. (See footnote 41 in Section 6 for an example of such a correspondence).

We may note that a constant correspondence Λ may be identified with a class of game forms.

In fact, throughout the present paper, when we speak of institutions as represented by classes $\mathcal{G}$ of game forms, they should be interpreted as correspondences Λ . (Alternatively, one can interpret such references as dealing with the special case of constant Λ .)

For the sake of brevity we sometimes refer to *conditional game forms* or *classes* $\mathcal{G}$ when the institutional aspects are represented by functions or correspondences such as Λ above. (The case of constant Λ may then be termed an *unconditional* game form.)

2.7.2 An alternative formalization of dependence on the environment

As an alternative way of looking at the problem, consider again the class $\mathcal{G}$ of conceivable game forms. Further, let $\mathcal{G}(e)$, a subset of $\mathcal{G}$, be the class of game forms consistent with the laws of physics and other restrictions that cannot be modified by human action under the prevailing environment e . Thus human action can only result in choice among the game forms in $\mathcal{G}(e)$. Thus an institution, say T , is represented by a subset, say G_T , of $\mathcal{G}(e)$.

In our interpretation of institutions as classes of game forms, the set-theoretic difference $\mathcal{G}(e) \setminus G_T$ represents the game forms that have been eliminated by human action. Full information about the effect of human action in this case is represented by

the ordered pair $\langle G(e), G_T \rangle$, showing both the field of choice and the choice made through human action.

Now the set $G(e)$ is determined by the environment e . If the correspondence $G(\cdot)$ is regarded as known, we are back to the ordered pairs (e, G) representing institutions, i.e., to conditional game forms.

3. The defining attributes of institutions

Not all classes of game forms qualify as institutions. In this section we shall be considering some of the attributes of game forms that, by almost unanimous consensus, characterize institutions. These are:

- a) The genesis of institutions: human actions (behaviour)
- b) Ensuring the effectiveness of institutions.
- c) The domain of applicability of institutional rules ("categoricity").

3.1 The genesis of institutions: human actions (behaviour)

Clearly, the constraints (or outcomes) due to laws of physics would not be considered institutions. In one fashion or another, this is recognized by all writers and incorporated into the definition of institutions. Thus, as seen from the above quotation, North defines institutions as *humanly devised* constraints.

Ostrom (1986, pp. 5–6) says: "Rules, as I wish to use the term, are distinct from physical and behavioral laws . . . That rules can be changed by humans is one of their key characteristics."

Schotter's definition (1981, p. 11) reads as follows:

A social institution is a regularity in social behavior that is agreed to by all members of society, specifies behavior in specific recurrent situations, and is either self-policed or policed by some external authority.

Clearly, it is the human behaviour or actions that generate the institutions. In Section 3.1.2 we consider a formalization (the concept of conditional game forms) designed to separate the "human design" from the "environmental" aspects of game forms.

The Reiter-Hughes (1981) model contains a sequence of two games, the first of which generates the (institutional) rules for the second. Again, human choices are involved. This formalization will be elaborated in Section 4 below.

3.1.1 Conscious design vs. 'organic' (endogenous) origin

It is to be noted that human actions are not synonymous with conscious design. While North speaks of "humanly devised" constraints, Schotter (1981, p. 21) stresses the absence of conscious design (the "organic" or 'behavioural' view, with roots in Carl Menger's (1883) discussion almost a century earlier):

. . . the institutional form will be an endogenous variable in the model. It emerges without any agent or group of agents consciously designing it – through human action but not human design.

In this context Schotter cites Hayek's (1955, p. 39) statement in a similar spirit:

The problems which they [the social sciences] try to answer arise only in so far as regularities are observed which are not the result of anybody's design.

The point that not all institutions emerge without conscious design is made in Schotter, although he gives reasons (1981, pp. 28–9) (some quoted from Hayek (1955)) why he will not (subsequently) be discussing consciously created institutions. He says:

... many social institutions are created in one stroke by a social planner or by the agents of society meeting in a face-to-face manner and bargaining about the type of institution they would like to see created. Here the exact form of the institution that emerges is the result of explicit human design (in the case of a planner) or multilateral bargaining (in the case of a legislature).¹⁹⁾

The “constitution creation game” is mentioned as an example of institutional rules created by such a process.

In his 1986 paper (pp. 117–118), Schotter correlates what he calls the “rules view” (“type I” in his 1989 terminology) of institutions with the notion of *consciously designed* game forms or mechanisms:

The focus of attention here is on the possibility of designing sets of rules or game forms (to use Gibbard's terminology) that when imposed on a set of social agents, lead to prespecified equilibrium outcomes. ... Throughout this literature, social institutions are planned and designed mechanisms given exogenously to or imposed upon a society of agents. Institutional change is a process of social engineering that takes place through the manipulation of rules.

But there is no logical basis for identifying the “rules view” of institutions with the assumption of conscious design as genesis of the rules. Many questions studied in what is often called “mechanism design” or “implementation”²⁰⁾ literature deal with the *logical* compatibility of various attributes of game forms or rules. The answers are as relevant to rules arrived at by consensus or majority vote of a group to whom the rules are to apply (and who respect the majority decision) as they are to rules “given exogenously or imposed upon a society of agents”.

This is, in particular, true of the informational (im)possibility theorems, such as the theorem stating that Pareto optimality cannot be guaranteed by any informationally decentralized mechanism with a message space of dimension substantially lower than that needed to verify that perfectly competitive equilibrium prevails (see Mount and Reiter (1974), Hurwicz (1977) and Osana (1978)). It is also true of the theorem stating that the set of Nash equilibrium allocations of any mechanism guaranteeing Pareto optimality in an economy where participants are not forced to trade must include the perfectly competitive allocations.²¹⁾

I see no reason why a set of rules under which a society or economy operates cannot

19) In fact, the first of two games in the Reiter-Hughes model, the (cooperative) political game Π [whose outcome (p. 1399) is the framework for the second game (played by firms and regulators)], is envisioned as a bargaining game involving coalition formation.

20) I probably am to blame for having popularized the “design” term, while “implementation” is probably due to Eric Maskin.

21) Subject to a condition of continuity and a sufficiently broad class of economies for which the guarantee is to hold. See Hurwicz (1979b).

be the outcome of some “organic” process, free of conscious design. In fact, I believe that Schotter’s (1981) supergame Nash equilibria may be viewed as an illustration of this possibility, showing that a rigorous game-theoretic formulation can constitute a model of “organic” or endogenous formation of rules governing behaviour. It is for this reason that I see no conflict between the rules interpretation of institutional arrangements and the possibility of “organic” genesis of such institutions.

However, the distinction between imposed and non-imposed rules does suggest a related and important dichotomy of institutions: (1. “external”) those, like rules contained in laws, that are binding on persons other than their creators (e.g., legislators), and (2. “internal”) those, like rules contained in contracts, that only bind those that participate in their creation (including heirs, assigns, etc.).²²⁾ My intent is to formulate the concept of institution so that it covers both categories. Moreover, if I think of the process of creating an institution as a game (à la Reiter-Hughes, and perhaps even Schotter), this is not meant to exclude the possibility of less formalized genesis such as custom formation.

3.2 Ensuring the effectiveness of institutions

Although Schotter’s concept of an institution differs from that proposed in the present paper²³⁾, it is valuable because of its explicit reference (1981, p. 11) to behaviour being “either self-policed or policed by some external authority”. North’s definition does not mention enforcement but its importance is clear from subsequent discussion (e.g., 1990, pp. 28–33). Crawford and Ostrom (1994, p. 6) distinguish rules from norms by the role of sanctions for not following rules (their “or else” element).

Schotter’s notion of “self-policed” may cover two phenomena: (1) behaviour being policed by members of the group rather than an external authority²⁴⁾, and (2) internalization of appropriate behaviour, so that each person is, in effect policing

22) It is interesting to note the part of Schotter’s definition of a social institution as “... a regularity of social behavior that is agreed to by all members of society ...” Taken literally, this would imply that all social institutions belong to class (2) above. The models of institution generation constructed by Schotter, including property rights, do satisfy this requirement. But one might not want to say that the institution of property rights does not exist if some members of the society are opposed to it.

23) Schotter (1981, p. 11) defines an institution as regularity of behaviour and calls it in Schotter (1989, pp. 50–51) an institution of type II, while labelling institutions defined in terms of rules as institutions of type I. Both type I and II are institutions in the B sense. (The terminology used in Crawford and Ostrom (1994) is “institutions-as-equilibria” and “institutions-as-rules” corresponding, at least roughly, to Schotter’s type II and type I respectively.)

In the terminology of game theory, an institution of type II is an equilibrium strategy (e.g., a Nash equilibrium) of a game, hence an element of the space S , while an institution of type I is formulated in terms of the game-form which, together with the players’ utility functions, defines a game.

As will be seen below, we take into account the equilibrium aspect of institutional phenomena by following Reiter and Hughes (1981) using a model consisting of a sequence of two games where the rules governing the second game are obtained as the equilibrium outcome of the first game. Thus an institutional arrangement is both a class of game-forms (of the second game) and an equilibrium outcome (of the first game). Note, however, that—except in special cases—the equilibrium outcome of the first game need not be an element of its strategy space.

24) A situation studied by Elinor Ostrom and co-authors in a number of papers.

himself/herself. In a variety of situations, any of these modes of enforcement (or their combinations) may be present. But it seems to me that enforcement is too narrow a category of phenomena serving to ensure the effectiveness of an institution (viewed as a class of game forms).²⁵⁾

Enforcement is aimed at making sure that the strategy domains prescribed by the game forms are being abided by. But to make a game form (mechanism) effective, the results specified by the outcome functions must also be “delivered”. Mechanisms such as markets or social insurance, illustrate the need for special machinery required to carry out, in addition to enforcement, the informational functions (in particular, communication) as well as the physical flow of goods and financial instruments. The totality of these required activities corresponds, I believe, fairly closely to everyday usage of the term “implementation”. But since in recent literature this term has been given a different meaning²⁶⁾, I sometimes qualify it by an adjective and speak of “*genuine implementation*”, with the intention of covering the complex of all activities designed to make the outcome function effective, thus including much in addition to enforcement.²⁷⁾

3.3 The domain of applicability of institutional rules (“categoricity”)

Here again, Schotter’s definition has the merit of explicitly referring to social institutions being concerned with “behaviour in specific recurrent situations”. To put it in a somewhat more general form, a (B-) institution deals with a *category* of situations (whether recurrent or contemporaneous) and typically applies to a class (*category*) of actors rather than to specifically named persons.²⁸⁾ Thus there are rules for buyers or sellers, for parents and children, for legislators, bureaucrats, etc. This is the property referred to as “*categoricity*”.

To determine that an institution applies in a given case, it must be verified that the situation and the actors belong to appropriate categories.

25) One occasionally hears the opinion that a Nash equilibrium is self-enforcing. But that is only true (presumably, by definition) if one assumes that players abide by the rules of the game. To make sure that they do, enforcement machinery may be necessary.

Another view exists, to the effect that enforcement is impossible because of the possibility of corruption on the part of those in charge of enforcement, as well as their superiors. (“Who will guard the guardians?”) An infinite regress seems to be a possibility. A framework for avoiding this paradox, involving the notion of the “natural game form” is discussed in Hurwicz (1993a).

26) Following Eric Maskin, current literature uses the expression “to implement a social choice rule” in the sense of finding a game form which, within a specified class of players’ preferences, would produce equilibrium outcomes consonant with the desiderata embodied in the social choice rule. (The use of the term “rule” in “social choice rule” to represent the desiderata, although widespread, is somewhat confusing in the present context; hence we sometimes use “goal correspondence” instead.)

27) One frequent aspect of genuine implementation is the creation of an agency (e.g., the Social Security Administration, an A-institution) whose function is precisely to make the outcome function operational, and in some cases also to enforce the constraints on the strategy domains.

28) As indicated above, some institutions (called above “internal”) govern only their creators; others (“external”) are binding on others as well.

4. A game sequence model

4.1 Three-game sequence model

To avoid ambiguity, we shall formalize some of the ideas discussed above. The formalization admittedly is narrower than the general situations discussed above; in particular, it may not cover the “organically” generated institutions that constitute Schotter’s primary interest. Specifically, we follow to a considerable extent the schema used by Reiter and Hughes, although we shall use a sequence of three games where Reiter and Hughes use two. It is the first of these three games, Γ_1 , called *preliminary*, that results in creating the institution, i.e., a class of game forms (denoted by $\mathbb{G}_{23} = \mathbb{G}_2 \times \mathbb{G}_3$) intended to govern the subsequent games.²⁹⁾³⁰⁾ The second game, Γ_2 , called *administrative* (although it may involve judicial as well as administrative bodies as players), formulates detailed regulations and so narrows down the class $\mathbb{G}_3$ to a single game form G_3 , required to be an element of $\mathbb{G}_3$. This game form G_3 will then govern the third game, Γ_3 , called *substantive*.

For example, the preliminary game Γ_1 may be the legislative process resulting in the prohibition of cartels, the institution being created; this legislation would define a class of game forms $\mathbb{G}_{23} = \mathbb{G}_2 \times \mathbb{G}_3$, assuming a law abiding society, one can expect that the natural game forms of the two subsequent games change appropriately, thus assuring compliance. Based on this legislation, in the course of the administrative game Γ_2 , the administrative (and/or judicial) bodies then formulate detailed regulations, e.g., defining the term “cartel” and specifying implementation and enforcement procedures consistent with the enabling legislation, thus producing a game form G_3 (an element of the class $\mathbb{G}_3$). Again, we shall assume a law abiding society, so that game form G_3 will in fact be effective in governing the substantive game Γ_3 . This game consists of interactions between the regulators and the firms being regulated, as well as among the firms themselves. (If compliance with law could not be taken for granted, additional actions would have to be taken in the course of the first two games to assure the effectiveness of the prohibition of cartels.) Thus, whatever the details, for the (desired) game form $G_3 = (S_3, h_3)$, the strategy domain S_3 excludes any ‘legal’ strategies that would result in the formation of a cartel among the participating firms, and h_3 imposes penalties for violations.

The question may be raised why a sequence of three separate games, is considered preferable to a single game with three stages. One reason is that the preliminary, administrative, and substantive phases are very likely to involve different sets of players. On the other hand for “internal” institutions a single multi-stage game model may be more appropriate.³¹⁾ Much of our formulation could be adapted to such a model.

29) Unlike Reiter and Hughes, however, we do not require at this stage that the preliminary game be a cooperative game.

30) Here $\mathbb{G}_2$ is assumed to be a one-element class (a singleton, its element denoted by G_2) representing the rules governing the administrative game Γ_2 . (Otherwise, a longer cascade might be needed.) On the other hand, $\mathbb{G}_3$ will in general have many elements, thus creating the need for the administrative stage, whose function is to narrow down the class $\mathbb{G}_3$ to a single element G_3 of this class that will govern the substantive game Γ_3 .

31) But even for “internal” institutions the sets of players in different phases may vary, as when, e.g., heirs of the contracting parties are involved.

The fact that we restrict ourselves to a sequence of just three-game models is mainly an expository simplification. In general, we might consider a ‘cascade’ of T games, $\Gamma_1, \dots, \Gamma_T$, $T \geq 2$, where each predecessor restricts the game forms governing its successors, and it is only the last game, Γ_T , is substantive. As an illustration, the sequence could consist of the country’s constitution, statutes of successively greater specificity, judicial interpretations of the law, as well as regulatory actions of administrative bodies.

Let us return to the three-game ($T = 3$) sequence model. For $t = 1, 2, 3$, let the set of players of Γ_t be $N_t = \{(t, 1), \dots, (t, n_t)\}$, the (joint) strategy space S_t , the outcome space Z_t , and the game-form $G_t = (S_t, h_t)$. (The symbol (t, i) denotes the i -th player in game Γ_t .) In game Γ_t , let the payoff function of player (t, i) be denoted by π_t^i . (It is obtained as the composition $\pi_t^i = \phi_t^i \circ h_t$, where ϕ_t^i is the utility function of player (t, i) and h_t the outcome function of Γ_t .) Then we may write $\Gamma_t = \langle N_t, S_t, (\pi_t^1, \dots, \pi_t^{n_t}) \rangle$, or, for short, $\Gamma_t = \langle N_t, S_t, \pi_t \rangle$ where π_t is the n_t -tuple of payoff functions of the game Γ_t .

The generic element z_1 of Z_1 (the outcome space of the preliminary game) is assumed to consist of two elements, say $z_1 = (G_{23}, b_{23})$, $G_{23} = G_2 \times G_3$, with G_2 consisting of the one element G_2 . Here G_3 is a class of “desired game forms” for the substantive game Γ_3 . A rough interpretation of this concept is that players of the preliminary game want the substantive game Γ_3 to be governed by one of the game-forms in G_3 .³²⁾ The b_{23} component of z_1 represents actions or behaviour of preliminary game players intended to have this effect on the environment prevailing during the subsequent games and hence the ‘natural’ game-forms $G^{\wedge}_2$ and $G^{\wedge}_3$ ³³⁾ of the administrative and substantive games as to make some game-form from G_2 effective for

32) More precisely, it is the result of strategy choices (bargaining, etc.) by members of N_1 each of whom is ‘pushing’ toward his/her favourite game-form for Γ_2 . Of course the equilibrium outcome G_2 need not be to every one’s liking.

33) Write the ‘natural’ game-form as $G^{\wedge} = (S^{\wedge}, h^{\wedge})$. The (“natural”) strategy domain $S^{\wedge}$ of $G^{\wedge}$ contains not only the strategies permitted by a desired game-form, but also all others that are physically and psychologically feasible. If we think of the desired strategies as the legal ones, $S^{\wedge}$ contains not only the legal strategies but also all feasible illegal strategies. By definition, the (“natural”) outcome function $h^{\wedge}$ of $G^{\wedge}$ represents those strategy choices from $S^{\wedge}$ that are in fact likely to occur. In general, even when desired strategies are used, the natural outcomes may differ from those desired. (See Hurwicz (1993a) for a fuller discussion of these concepts.)

Consider games with two players and finite strategy spaces, so that the outcome functions are represented by matrices. We may want to suppose that the desired outcome matrix is a submatrix of the natural outcome matrix, i.e., that at least the legal strategies have the desired outcomes. However, this need not always be so. It may easily happen that the desired outcome function is not effective, so that the actual (natural) outcome associated with a “legal” joint strategy s is different from the prescribed outcome; i.e., in symbols, for some $s \in S$, $h_{\wedge}(s) \neq h(s)$ where $h^{\wedge}$ denotes the natural outcome matrix and h the desired outcome matrix; thus in this case the desired outcome matrix h is not a submatrix of $h^{\wedge}$.

A desired game-form $G^* = (S^*, h^*)$ is said to be *effective* with respect to the natural game-form $G^{\wedge}$ over a class Φ of n -tuples of utility functions for players in N if, for every n -tuple of utility functions ϕ in Φ , the sets of equilibrium outcomes of the two games $\Gamma^{\wedge}$ and G^* are the same. (Here $\Gamma^{\wedge}$ is the composition of ϕ with the outcome function $h^{\wedge}$, while Γ^* is the composition of the same ϕ with the outcome function h^* ; the set of players is N in both cases.)

The intent of this definition of the team “effective” is to formalize the idea that even though the actual game being played is always the natural game, the outcomes will be the same as those that would have resulted from the desired game. In particular, even though participants could have used illegal strategies, they will not do so at an equilibrium. Thus enforcement works and we have “genuine implementation”.

Γ_2 and some game form from $\mathbb{G}_3$ effective for Γ_3 . The introduction of the element b_{23} is motivated by the idea that the desired game-forms in $\mathbb{G}_2$ might remain “on paper” and not become effective unless the natural game form of the substantive game was suitably changed by some action or behaviour. Examples of b_{23} components are: physical barriers (preventing or deterring the use of “illegal” strategies), education and propaganda, and, of course, the existence or creation of reliable and effective implementation (in particular, enforcement) organs.³⁴ These activities would involve diversion of resources (a component of transaction costs). A more sophisticated (probabilistic) model would explicitly take into account the fact that enforcement typically is less than 100% successful. It is interesting to note that in experimental settings great effort is made to make the rules of the game completely effective (there is genuine implementation), often by the nature of physical arrangements.

The outcome z_2 of G_2 is of the form (G^*_3, b_3) , where G^*_3 is the game form (no longer a class!) designed to govern the substantive game Γ_3 and b^*_3 is again an action aimed at modifying the environment E_3 and hence the natural game form $G^{\wedge}_3$ prevailing during the substantive game so as to assure effectiveness.

In certain situations or societies, the mere fact that the preliminary game had produced the class $\mathbb{G}_{23}$ might have sufficient authority to make one of its constituent game-forms effective. (We speak of law-abiding societies and say that the rule of law prevails.) In such cases, b_{23} might be interpreted simply as the announcement of the class $\mathbb{G}_{23}$ as the equilibrium outcome of the preliminary game. (Formally, in this case, $z_1 = (\mathbb{G}_{23}, b_{23})$, with b_{23} = the announcement of $\mathbb{G}_{23}$, and similarly b^*_3 = the announcement of G^*_3 .)

There is an important relationship between preferences of the players, N_1 and N_3 , respectively in the preliminary and substantive games of the sequence. Typically, a player of the preliminary game is interested in the expected or likely outcome z_3 of the substantive game, perhaps because he/she identifies with the interests and/or preferences of some players of the substantive game. Hence while the utility function ϕ_1^i has Z_1 as its domain, the value of $\phi_1^i(z_1)$ depends on the expected outcome z_3 generated (indirectly) by z_1 .

4.2 Main features of the model

To summarize, among the important features of the preceding model are these: (a) the sequential structure of the process; (b) the preliminary game, if effective (i.e., genuinely implemented), determines a class of game forms, here interpreted as the institutional arrangements, that will ultimately govern the “proper” phase of the substantive game; (c) in order to achieve genuine implementation, it is, in general, necessary for the preliminary game to produce actions or behaviour that will so modify the natural game form of the substantive game as to make the rules produced by the preliminary game effective.³⁵

It should perhaps be emphasized that the model is intended for a broad class of institutions, even though in many cases it would not be natural to represent the

34) See Hurwicz (1993a) on the role of “intervenor”, and “closed circles”.

35) A formalization of “effective” is given above in footnote 33.

formative phase by game formalism. However, other aspects of the model might still be appropriate. Thus, for instance, the common features of the institution of marriage, with its many forms in many societies and historical periods, may be represented by a class $\mathbb{G}_3$ of game forms, while the formative phase would be represented by an “organic” evolutionary process. However, the specific game form G^*_3 prevailing in a country or state during a particular period (e.g., the prohibition of polygamy) often is the product of legislative, administrative, or judicial actions.

As for the internal structure of the substantive game, whether or not it includes the administrative phase, it may be most natural to represent it in extensive form. One reason is that in practice, rules are usually³⁶⁾ formulated in terms of moves rather than strategies. Moreover, extensive form would make it simpler to accommodate an important aspect of New Institutional Economics, viz. transaction costs. Therefore, for purposes of institutional analysis, it would be desirable to introduce the notion of *game-form in extensive form*. This could be represented by the usual tree diagram, but with outcomes (elements of the outcome space Z) instead of numerical (utility) payoffs. Furthermore, it would also seem desirable to depart from the usual procedure of associating the outcomes with terminal nodes only. In addition to the usual terminal outcomes, one could introduce *interim outcomes*, associated with progression from any node to the next. The various transaction costs would be among the components of such interim outcomes. An example might be the requirement of up-front payments for the filing of a law-suit.³⁷⁾

4.3 Definition of an institution

We have so far discussed different characteristics and models of (B-)institutions, but no formal definition of an institution has been given. A highly simplified definition might be that implied by the title of this paper: an institution is a class of game forms. In the preceding example of the institution prohibiting cartels, it would be a class $\mathbb{G}_3$ of various game forms all of which exclude strategy domains involving cartel formation and, perhaps, some limits on penalties for violations. But such a definition fails to embody the various attributes that qualify a game-form class as an institution. To begin with, in line with our discussion in Section 2.7 above, in order to make sure that we are excluding the effects of the laws of physics, etc., we must use conditional game forms (in effect, specifying classes of game forms permitted by human actions given the prevailing environment). Second, there must be explicit reference to human actions creating the institution, the multi-stage or sequential aspect of the process. Third, since effectiveness (or genuine implementation) is considered to be essential, the element making for effectiveness must also be explicit in the formalization. Finally, there must be a formal expression of the categoricity aspect.

Clearly, this makes for a complex concept. In the interest of minimizing this complexity, we shall confine ourselves to the sequential structure described in Section

36) But not always: for instance, rules dealing with predatory competition concern strategies rather than separate moves.

37) Introducing interim outcomes is not a complete innovation. In games with infinite time horizon there are, of course, no terminal nodes; the so-called felicities are interim utilities associated with interim outcomes such as consumption in a given time period.

4.1. Formally then, a (B-)institution is defined as the following ordered sequence of elements:

$$\langle E_1, \Gamma_1, z_1; E_2, \Gamma_2, z_2; E_3 \rangle,$$

with

$$z_1 = (\mathbb{G}_{23}, b_{23}), \mathbb{G}_{12} = \mathbb{G}_2 \times \mathbb{G}_3, z_2 = (\mathbb{G}_3^*, b_3^*),$$

$$E_2 = \eta_2(E_1, b_{23}); E_3 = \eta_3(E_2; b_{23}, b_3^*)$$

all game forms being “conditional” (in the sense of Section 2.7)³⁸⁾ and “categorical” (in the sense of Section 3.3, i.e., applicable to categories of actors and situations rather than named individuals and situations), the unique element G_2 of $\mathbb{G}_2$ being effective in Γ_2 , and the game form G_3^* being effective for the class $\mathbb{G}_3$ over E_3 .³⁹⁾

The components b_{23} and b_3^* represent actions or behaviours so modifying the environment as to assure effectiveness in subsequent games. E_t represents the environment prevailing during game Γ_t and hence determines the natural game form G_{Λ_t} prevailing during Γ_t . The symbols η^2 and η^3 denote functional relations representing the effects of the respective actions b_{23} and b_3^* on subsequent environments.

As it stands, the preceding formulation does not accommodate institutions whose “organic” genesis does not lend itself to game-theoretic representation of the preliminary phase ($t = 1$). To model this type of genesis, the preliminary game Γ_1 would have to be replaced by some non-game type of process. However, other elements of the above formal definition might still retain their relevance.

On the other hand there is a relationship, albeit imperfect, between Schotter’s supergame model of institutions and our above framework, with the supergame corresponding to our preliminary game Γ_1 , and the solution of the supergame corresponding to our game form G_3^* .

5. Contents of institutions

So far we have been discussing the more formal aspects of modelling institutions. But what of their substance? In particular, in economic settings, what aspects of the economic process do the rules deal with? The answer, of course, is: a tremendous diversity. Without attempting a complete treatment, it seems worthwhile to indicate two types of phenomena frequently treated in institutional literature: endowments and residuals.

38) Hence, strictly speaking, we should be using the symbols Λ representing correspondences from the respective environment spaces into the space of conceivable game forms rather than the symbols $\mathbb{G}$ representing classes of game forms. However, as indicated in Section 2.7, we have opted for the somewhat ambiguous notation because the $\mathbb{G}$ symbols seem more suggestive, and, of course, correct in the case when the Λ ’s are constant.

39) It may be noted that the class of environments, E_3 over which the institution represented by $\mathbb{G}_3$ is effective may be different from what had been aimed at by players in the preliminary game.

5.1 Institutions assigning initial endowments

A broad class of institutions dealt with by economists can be formalized in terms of the assignment of initial *endowments*, especially endowments of rights (see Eggertsson (1990), especially ch. 4). Among many examples are the following.

The alternative liability rules for damage caused by pollution can be represented by alternative assignments of initial endowments of the right to pollute or the right to enjoy clean air and water (see Hurwicz (1993c)).

The distinction between common and (individual) private property (central to the “tragedy of the commons” problem) can be represented by distinguishing between endowments associated with a group of agents as against endowments obtained by individual agents. A property right may be viewed as a ternary relation involving a person, natural or artificial, the specific object owned, and a particular right, e.g., that of use, or alienation. Specifically, consider a particular aspect of property rights, the right of use, say the right of fishing in a commonly owned body of water. In general, more than one person may have the right of fishing there. But under exclusive individual property rights, only one person has such right. Denote by (i, x) the right of agent i to use the object x . In game-form formulation this means that (i, x) is an element of i 's strategy domain S^i . The absence of common ownership means that, for any $j \neq i$, the element (j, x) does not belong to S^j . So absence of common ownership is a relationship between the different players' strategic domains, hence an attribute of the (joint) strategy space S , and therefore of the game form (S, h) .

Primogeniture, considered by many an important institutional factor in England's economic history, is naturally represented as a rule concerning the endowment of heirs with the parents' estate. (Note that the rule is of “categorical” nature: it applies to the actor defined as the oldest son, not to a specific person.) In some cases the rights endowments are of procedural nature (e.g., the burden of proof in torts⁴⁰), materially affecting transaction costs.

5.2 Institutions assigning certain residuals

Another class of institutional phenomena deals with certain *residuals*. Thus Grossman and Hart (1986) explain the effects of vertical or lateral integration of firms by the residual rights of ownership with respect to vague residuals due to the incompleteness of contracts caused by unavoidable complexity of contingencies. The different incentive effects of the various land tenure forms are due, at least in part, to the location of residual risks or opportunities. Similar considerations apply to alternative institutional structures of firms, both with respect to rewards (profits, etc.) and liabilities. The location of these residuals is often determined by the assignment of endowments, including those of liabilities. Hence there is a relationship between the two aspects of institutional phenomena, and much of the principal-agent literature deals with them. Very tentative and incomplete ideas for certain aspects of these phenomena are discussed in the next section.

40) In the Napoleonic code, as well as in ancient Roman Law, when damage was due to actions of children, cattle, or was “caused” by buildings, the burden of proof was on parent or owner, as an exception to the general rule. In French civil law before World War II this reversal also applied to most cases of damage caused by automobile accidents.

5.3 Institutions governing particular aspects of behaviour

Institutions are not only due to human actions, they also deal with human actions, whether through the description of the strategy domains S^i or through the specified outcome functions h . One problem of formalization we have not yet faced is the fact that what is usually called an institution deals only with particular aspects of human activities, e.g., commerce, family status, punishment for crimes, etc., rather than with the totality of human behaviour. To formalize this, consider again an institution T (say, rules governing commerce) represented by the class G_T of game forms. Let the outcome space Z be the common range of all outcome functions in G_T , and let us think (in the present context only) of Z as a subset of a vector space, with sets of components of its elements corresponding to different aspects of human behaviour. For such an element z , write $z = (z_T, z_{T'})$, an element of the Cartesian product $Z = Z_T \times Z_{T'}$. Here z_T represents the components dealing with behaviour in commerce, while $z_{T'}$ represents all other aspects of behaviour. If $(S\#, h\#)$ is an element of G_T , and if, for some s in $S\#$, $h\#(s) = (a^*, b^*)$ in $Z_T \times Z_{T'}$, then G_T must also contain all game forms $(S\#, h)$ where $h(s) = (a^*, b)$ and b ranges over all of $Z_{T'}$.

Similar factoring of the elements s of the strategy domains S is also to be carried out. Thus if s is in S , write $s = (s_T, s_{T'})$, an element of the Cartesian product $S = S_T \times S_{T'}$. Then if $(S\#, h\#)$ is an element of G_T , and $s^* = (c^*, d^*)$ is in $S\#$, the G_T must also contain all game forms $(S, h\#)$ where S contains all elements of the form $s = (c^*, d)$, with d ranging over all of $S_{T'}$.

Let the different aspects of behaviour be indexed by the set Θ , and let the family G_θ of game forms represent the institution dealing with aspect θ of behaviour. Then, clearly, the aggregate impact of the totality of institutions G_θ , with θ ranging over Θ , is represented by their intersection $\bigcap_{\theta \in \Theta} G_\theta$. If the requirements of the various institutions are not mutually inconsistent, the intersection is non-empty.

6. The characteristic function, the support set, and the support function

As an illustration of some of the preceding concepts consider the following situation, closely related to the principal-agent model, with the Stiglitz land tenure example of Section 2.6 and footnote 17 as an illustration.

There are two persons 1, 2, with strategy domains S^1 and S^2 . The set S^0 is the strategy domain of the exogenous factors (e.g., weather). S^i includes effort expended by person i , $i = 1, 2$. The joint strategy space is $S = S^1 \times S^2 \times S^0$.

Write $s = (s^1, s^2, s^0)$ for the generic element of S .

The generic outcome is written

$$z = (y, r^1, r^2),$$

where y is the output (or, e.g., value added) and r^i the reward (wealth increment) for person i .

The production function is

$$y = \eta(s);$$

in particular, the function η represents the extent to which the output depends on the level of effort of the two persons and on the exogenous factors. (The symbol η used here is not related to the same symbol used in Section 4.)

The reward functions f^i (defined by the prevailing institutions) are given by

$$r^i = f^i(y; s), \quad i = 1, 2.$$

(When the strategy, e.g., the level of effort, of person i is not observable to the other, it may be impractical to make the reward function f^i directly dependent on i 's effort or, more generally, on s^i . A similar limitation applies to direct dependence of reward functions on exogenous factors, i.e., on s^0 .)

Hence the outcome function $h(\cdot)$ can be written as

$$h(s) = \langle \eta(s), (f^i(\eta(s); s))_{i=1, 2} \rangle. \quad (41) \quad (*)$$

The utility functions are $u^i(r^i, s)$, $i = 1, 2$. It is natural to assume that u^i rises with r^i . The dependence of u^i on s represents phenomena such as the disutility of effort, the discomforts of bad weather, etc.

The production function η tells us who and to what extent strategies, in particular effort, can affect the output. On the other hand, the reward function f^i tells us to what extent i 's reward is affected by variations in output, as well as directly by s . The relationship between the reward functions and the production function is a major factor in determining whether the given institutional setup does or does not promote "efficiency" (as measured by output).

The situation can be analysed with the help of certain concepts that seem of potential value in arriving at general propositions about the incentive structures engendered by alternative institutional arrangements.

The first of these is what we call the *characteristic function of an institution*, denoted by F . For instance, consider the Stiglitz land tenure example of Section 2.6 and footnote 17. Let the three forms of land tenure be abbreviated as a = wages, b = renting, c = sharecropping, and let the two persons be the landlord ($i = 1$) and the worker ($i = 2$). Then we define $F(x/i) = 1$ (or -1 , or 0) if i 's reward depends positively (or negatively, or does not depend) on i 's s^i (e.g., effort) via the production function.

Thus, assuming that the worker's effort can increase output, $F(2/a) = 0$ while $F(2/b) = 1$. On the other hand, if the landlord's actions can increase the worker's productivity, one might expect that $F(1/a) = 1$ while $F(1/b) = 0$. Finally, if both sharecroppers can raise output by their effort, $F(1/c) = F(2/c) = 1$.

The *support set of an institution* x is defined as the set of agents i , denoted by $\psi(x)$, such that $F(i/x) \neq 0$. The *support function for a class X of institutions* (with the same set of agents N) is the set-valued function $\psi(\cdot)$ where $\psi(x)$ is the support set of x , and the argument of $\psi(\cdot)$ ranges over the class X . I.e., $\psi: X \rightarrow N$, where for a given institution x in X , $\psi(x) = \{N' \text{ a subset of } N: F(i/x) \neq 0 \text{ for } i \text{ in } N'\}$.

Final version accepted November 24, 1995

41) Our interpretation here is that only the reward functions represent the institutional aspect of the model while the production function is an aspect of the environment. Consider, for example the class L of reward functions affine in y , i.e., $f = (f^1, f^2)$, $f^2(y, s) = ky + b$, $f^1(y, s) = y - f^2(y, s)$ where the landlord is agent 1, and let the subset L^* of L be the class of reward functions corresponding to sharecropping, i.e., where $f^2(y, s) = ky$, with k ranging over the open interval $(0, 1)$. Then the institution of sharecropping is formally represented by the correspondence $\Lambda: E \rightarrow \mathbb{G}$ where an element e of the space E of environments can be identified with a particular production functions η and $\mathbb{G}$ is a class of game forms, and $\Lambda(e) = \Lambda(\eta) = h(s)$ where h is given by the formula $(*)$ in the text. (See Section 2.7 for a discussion of conditional game forms.)

REFERENCES

- Coase, R. (1960) "The Problem of Social Cost," *Journal of Law and Economics*, Vol. 3, pp. 1–44.
- Crawford, S. and E. Ostrom (1994) "A Grammar of Institutions," 1994 [marked 7/8/94 and W93-1], Workshop in Political Theory and Policy Analysis, Indiana Univ., Bloomington, Indiana.
- Eggertsson, T. (1990) *Economic Behavior and Institutions*, Cambridge: Cambridge University Press.
- Gardner, R. and E. Ostrom (1991) "Rules and Games," *Public Choice*, Vol. 70, pp. 121–149.
- Grossman, S. J. and O. D. Hart (1986) "The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration," *Journal of Political Economy*, Vol. 94, pp. 691–719.
- Hayek, F. A. (1955) *The Counterrevolution of Science*, New York: New York Free Press.
- Hurwicz, Leonid (1972) "On Informationally Decentralized Systems," in McGuire, C. B. and R. Radner, eds., *Decision and Organization*, Amsterdam: North Holland, ch. 14.
- (1977) "On the Dimensional Requirements for Non-wasteful Resource Allocation Systems," in Arrow, K. J. and L. Hurwicz, eds., *Studies in Resource Allocation Processes*, Cambridge, Cambridge University Press.
- (1979a) "On the Interaction between Information and Incentives in Organizations," in Krippendorff, K., ed., *Communication and Control in Society*, New York: Gordon and Breach Science Publishers.
- (1979b) "On Allocations Attainable through Nash Equilibria," *Journal of Economic Theory*, Vol. 21, pp. 140–165.
- (1993a) "Implementation and Enforcement in Institutional Modelling," in Barnett, W. A., M. J. Hinich and N. J. Schofield, eds., *Political Economy: Institutions, Competition and Representation*, New York: Cambridge University Press, ch. 2.
- (1993b) "Toward a Framework for Analyzing Institutions and Institutional Change," in Bowles, S., H. Gintis and B. Gustafsson, eds., *Markets and Democracy: Participation, Accountability and Efficiency*, Cambridge: Cambridge University Press.
- (1993c) "What is the Coase Theorem?" presented at the Seventh Annual Japan-U.S. Technical Symposium, *The Coase Theorem: Competition and Bargaining, held on March 5, 1993, at New York University*; published in *Japan and the World Economy*, Vol. 7, (1995), pp. 49–74.
- (1994) "Institutional Change and the Theory of Mechanism Design," *Academia Economic Papers*, Vol. 22, No. 2, (September, 1994), Taipei, Taiwan: Institute of Economics, Academia Sinica.
- , E. Maskin and A. Postlewaite (1995) "Feasible Nash Implementation of Social Choice Rules when the Designer does not Know Endowments or Production Sets," in Ledyard, J. ed., *The Economics of International Decentralization: Complexity, Efficiency and Stability. Essays in Honor of Stanley Reiter*, Boston: Kluwer Academic Publishers.
- Marshall, A. (1890) *Principles of Economics*, Vol. 1, London: Macmillan.
- Menger, C. (1883) *Untersuchungen ueber die Methode der Sozialwissenschaften und der politischen Oekonomie insbesondere* (Investigation of the Methods of the Social Sciences especially Political Economy), Leipzig: Duncker and Humblot (reprinted by the London School of Economics, 1933).
- Mount, K. and S. Reiter (1974) "The Informational Size of Message Spaces," *Journal of Economic Theory*, Vol. 8, pp. 161–192.
- North, D. C. (1990) *Institutions, Institutional Change and Economic Performance*, Cambridge: Cambridge University Press.
- (1992) *Transaction Costs, Institutions, and Economic Performance*, San Francisco, CA: International Center for Economic Growth, ICS Press.
- Osana, H. (1978) "On the Informational Size of Message Spaces for Resource Allocation Processes," *Journal of Economic Theory*, Vol. 17, pp. 66–78.
- Ostrom, E. (1986) "An Agenda for the Study of Institutions," *Public Choice*, Vol. 48, pp. 3–25.
- , J. Walker, and R. Gardner (1992) "Covenants with and without a Sword: Self-governance is Possible," *American Political Science Review*, Vol. 86, No. 2, pp. 404–417.
- Reiter, S. and J. Hughes (1981) "A Preface on Modeling the Regulated U.S. Economy," *Hofstra Law Review*, Vol. 9, No. 5, pp. 1381–1421.
- Riker, W. H. (1982) "Implications from the Disequilibrium of Majority Rule for the Study of Institutions," in Ordeshook, P. C. and K. A. Shepsie, eds., *Political Equilibrium*, Boston: Kluwer-Nijhoff, pp. 3–24. Originally in *American Political Science Review*, 1980, pp. 432–447.
- Schotter, A. (1981) *The Economic Theory of Social Institutions*, Cambridge: Cambridge University Press.
- (1986) "The Evolution of Rules," in Langlois, R. N., ed., *Economics as a Process*, Cambridge: Cambridge University Press, ch. 5.
- (1989) "Comment" (on "Market and Institutions" by Siro Lombardini), in Shiraishi, T. and S. Tsuru, eds., *Economic Institutions in Dynamic Society*, London: Macmillan and International Economic Association.
- Stiglitz, J. (1974) "Incentives and Risk Sharing in Sharecropping," *Review of Economic Studies*, Vol. 41, pp. 219–256.
- von Neumann, J. and O. Morgenstern (1944) *Theory of Games and Economic Behavior*, Princeton: Princeton University Press.